

INVENTRA · MULTI-AGENT DESIGN

How a stockout case moves through Inventra

Three narrow-scope LLM agents hand a case to each other through a LangGraph state machine. A human approves once. Every database write happens in code the agents never touch.

LLM agent – reads facts, cannot invent them

☐ Deterministic node – plain code, no model call

Human approval gate – the one real pause

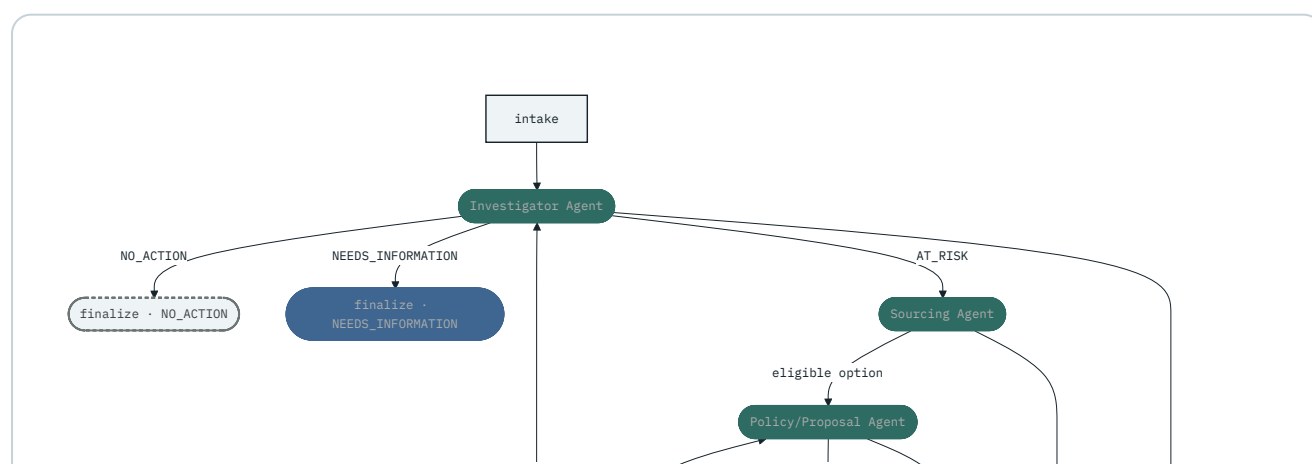
01 THE GRAPH 02 TRUST, BUT VERIFY 03 PAUSE & RESUME 04 AGENTS AT A GLANCE

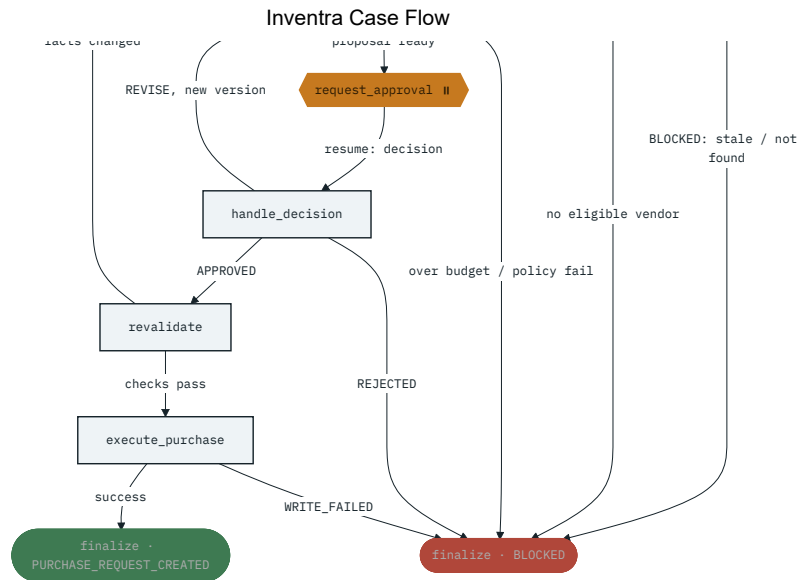
05 SCENARIO COVERAGE

01

The graph

One case enters at intake and can only reach a database write by passing through every agent, the human gate, and a revalidation check. Five different failure reasons all exit through the same `finalize · BLOCKED` terminal — only one path reaches `PURCHASE_REQUEST_CREATED`.





Claim: only the APPROVED path from a human, followed by a passing revalidate, can reach a write — every other branch, from every agent, dead-ends at a read-only BLOCKED or informational terminal.

02

Trust, but verify

Each agent's LLM call is bound to the same raw tool evidence as a plain Python function sitting right next to it. The function, not the model, decides the status that leaves the node.

RAW TOOL EVIDENCE

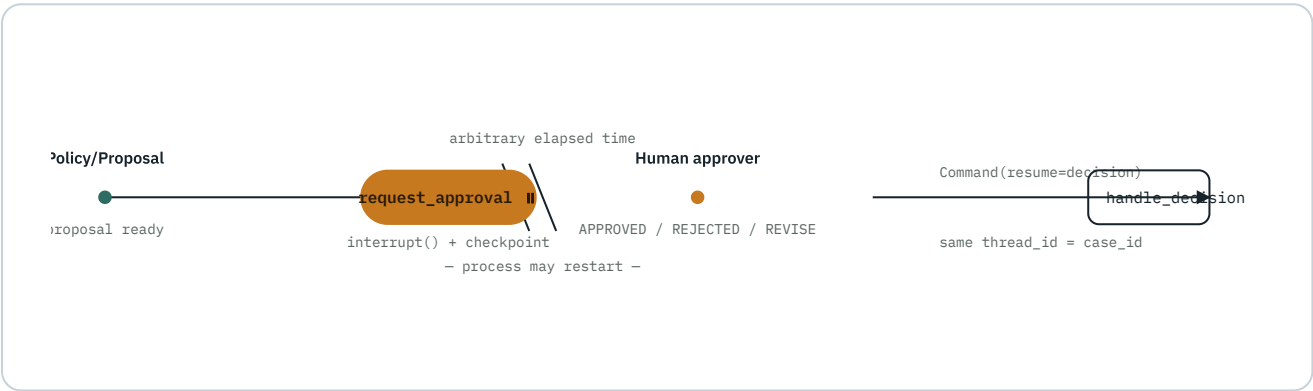


Claim: the LLM's proposed status never reaches InvestigatorOutput — `derive_status()` reads the same raw evidence independently and overwrites it. Only the LLM's narrative survives untouched. This pattern repeats for the Sourcing and Policy/Proposal agents.

03

Pause & resume

Approval is a durable interrupt, not a chat reply. The graph can sit paused for minutes or days — even across a process restart — because state is checkpointed under `case_id` as the LangGraph thread id.



Claim: nothing about resuming depends on wall-clock timing or an open connection —
`graph.invoke(Command(resume=...), {"configurable": {"thread_id": case_id}})`
re-enters cold at `handle_decision` with the full case state intact.

04

Agents at a glance

Each agent gets the smallest toolset that lets it do its one job, and none of them can call a write tool.

Investigator

BUILT

Decide whether a case needs sourcing at all — or ends right here.

get_product

get_stock_position

Sourcing

PLANNED

Find a grounded, policy-eligible vendor option for an at-risk case.

list_vendor_offers

get_sales_velocity

calculate_stock_risk

derive_status() recomputes NO_ACTION / NEEDS_INFORMATION / BLOCKED / AT_RISK straight from tool evidence.

get_vendor_performance

build_vendor_options

recommend_vendor_option

Picks a strategy and narrates cost-vs-speed; cannot invent a vendor, price, or lead time — those come from VendorOption objects, not text.

Policy / Proposal

PLANNED

Apply policy.md, draft the proposal, gate it on budget before a human sees it.

get_budget_position

get_policy_guidance

draft_replenishment_proposal

Budget check runs in code — the LLM can argue for a policy exception, but can't flip policy_passed against a hard budget failure.

05

Scenario coverage

All 12 acceptance scenarios map to a specific mechanism above, not a special case bolted on.

#	SCENARIO	OUTCOME	MECHANISM
1	Healthy stock	NO_ACTION	Investigator ends the case before Sourcing is ever entered
2	Missing request	NEEDS_INFO	validate_input() returns before any tool call

data			
3	Stale inventory snapshot	BLOCKED	derive_status() checks risk.stale
4	Cost vs. speed trade-off	AWAITING	Sourcing narrates recommend_vendor_option's trade-off
5	Over-budget proposal	BLOCKED	Policy node compares cost to budget.remaining in code
6	Invalid model output	BLOCKED	One repair re-invoke, then fail closed (<i>not yet implemented</i>)
7	Approval data changes mid-pause	BLOCKED	revalidate fails → routes back to Investigator
8	Duplicate approval resume	CREATED	Deterministic idempotency_key + DB unique constraint
9	Human rejection	BLOCKED	handle_decision on REJECTED, audit logs approver + comment
10	Database write failure	BLOCKED	create_purchase_request exception → WRITE_FAILED, no partial state
11	Insufficient sales history	NEEDS_INFO	sales.error == INSUFFICIENT_DATA short-circuits risk calc